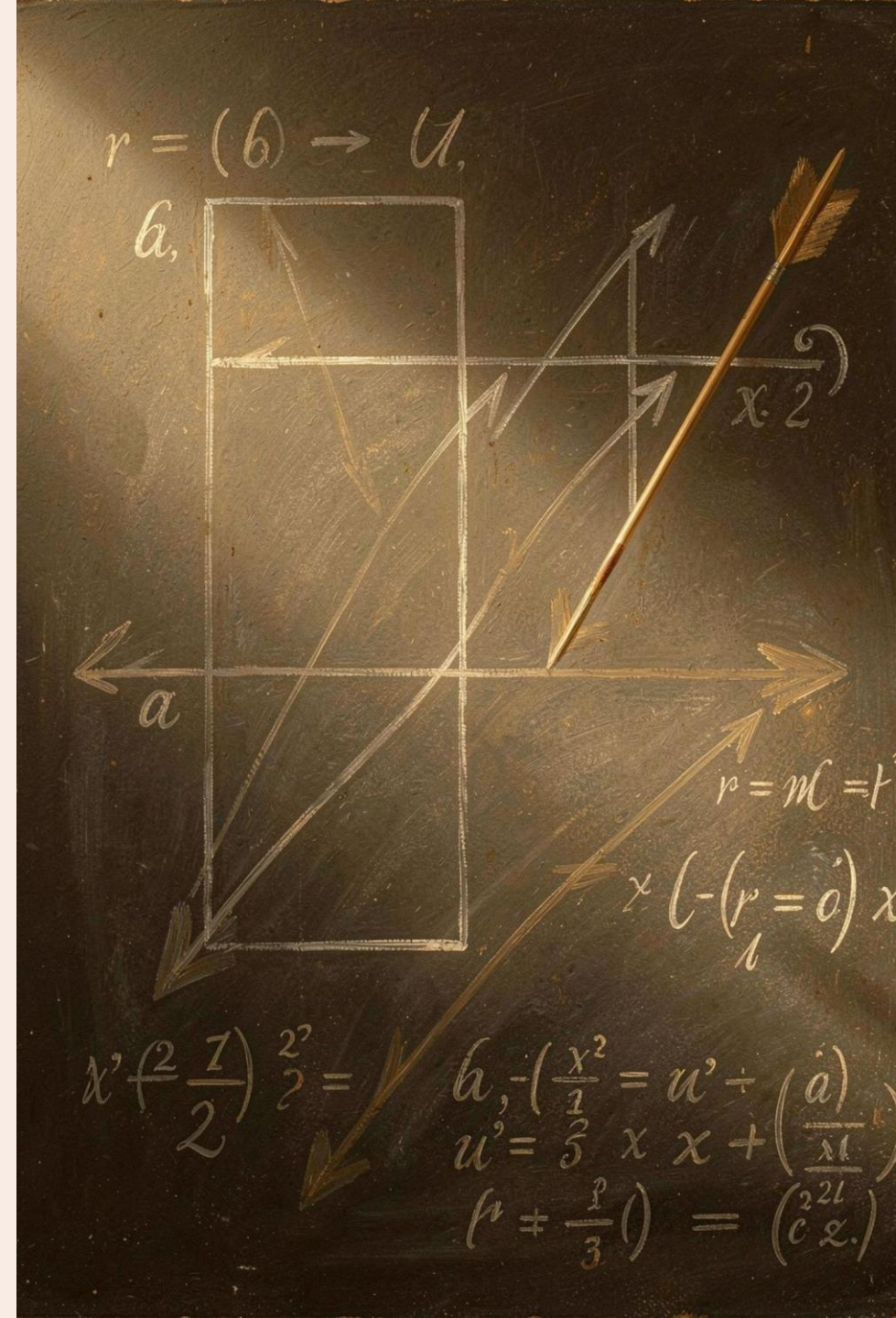


Basic Concepts of Vectors

Vector Analysis • B.Sc. Engineering Mathematics

ENGINEERING MATHEMATICS VECTOR ANALYSIS



Learning Objectives

By the end of this module, you should be able to confidently work with vector quantities in both mathematical and engineering contexts. These objectives form the foundation for everything that follows in vector analysis.

1	Define Scalar & Vector Quantities Understand the fundamental distinction between quantities described by magnitude alone versus those requiring both magnitude and direction.
2	Distinguish with Engineering Examples Identify real-world engineering scenarios — from structural forces to fluid flow — where scalar and vector quantities appear.
3	Identify Common Vector Types Recognize zero, unit, equal, opposite, parallel, collinear, and coplanar vectors and understand their geometric significance.
4	Calculate Dot and Cross product Understand and calculate both the scalar (dot) product and the vector (cross) product, interpreting their geometric and physical meanings.
5	Perform Basic Vector Operations Add, subtract, and scale vectors algebraically and interpret the geometric meaning of each operation.
6	Understand Geometric Meaning Visualize vector operations as geometric constructions.

What is a Vector?

A **vector** is a physical or mathematical quantity that possesses both **magnitude** (size or length) and **direction**. This dual nature is what distinguishes vectors from ordinary numbers.

Geometrically, a vector is represented by a **directed line segment** an arrow where the length encodes magnitude and the arrowhead indicates direction.

Definition: A vector is an object that has both a [magnitude](#) and a direction. A vector can be denoted as A , $\mathbf{A}$, $\vec{A}$, $\underline{A}$. Suppose, $\underline{A} = a\hat{i} + b\hat{j} + c\hat{k}$, be a vector in 3-D, where $\hat{i}, \hat{j}, \hat{k}$ are unit vectors along X, Y and Z-axis. We can also write it as $\vec{A} = (a, b, c)$.

Examples

→ Displacement

Change in position of an object requires both how far and in which direction.

→ Velocity

Rate of change of displacement speed alone is insufficient; direction matters.

→ Force

A push or pull acting on a body characterized by magnitude, direction, and point of application.

→ Momentum

Mass times velocity inherits the directional property of velocity.

Types of Vectors

Vector quantities can be classified into several types based on their magnitude, direction, and spatial relationships. Understanding these categories is essential for vector algebra and engineering applications.

Zero (Null) Vector

Magnitude is **zero**; direction is **undefined**.
Denoted as **0**. Represents no displacement or no net force.

Unit Vector

Magnitude is exactly **1**. Used to specify direction without regard to magnitude.
Denoted with $\hat{a}$.

Equal Vectors

Two vectors are equal if they have the **same magnitude** and the **same direction**, regardless of their positions.

Negative Vector

The negative of $\vec{A}$, written $-\vec{A}$, has the **same magnitude** but **opposite direction**.

Parallel Vectors

Vectors that lie along **parallel lines**. They may point in the same or opposite directions.

Collinear Vectors

Vectors that lie on the **same straight line**. All collinear vectors are parallel, but not all parallel vectors are collinear.

Coplanar Vectors

Vectors that all lie in the **same plane**. Any two vectors are always coplanar; three or more may or may not be.

Scalar vs. Vector Quantities

One of the most important distinctions in engineering mathematics is knowing whether a quantity is a scalar or a vector. Misidentifying the type can lead to serious errors in analysis and design.

Scalar Quantity

Described **completely by magnitude only**. No direction is associated with a scalar. Scalars follow the ordinary rules of algebra.

- Mass (kg)
- Time (s)
- Temperature ($^{\circ}\text{C}$ or K)
- Energy (J)
- Speed (m/s)
- Distance (m)
- Electric charge (C)

Vector Quantity

Requires **both magnitude and direction** for complete description. Vectors follow vector algebra, not ordinary arithmetic.

- Displacement (m)
- Velocity (m/s)
- Acceleration (m/s^2)
- Force (N)
- Momentum ($\text{kg}\cdot\text{m/s}$)
- Electric field (N/C)
- Weight (N)

Representation of a Vector

Geometric Representation

A vector from point **A** to point **B** is written as $\overrightarrow{AB}$. The point **A** is called the **tail** (initial point), and point **B** is called the **head** (terminal point). The arrow indicates the **direction** of the vector, and the length of the arrow represents the **magnitude** according to a chosen scale.

Two vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are equal if and only if they have **equal magnitude** and the **same direction** their positions in space do not need to coincide.

Algebraic Representation

In Cartesian coordinates, a vector is expressed using its components along the coordinate axes:

$$\mathbf{A} = A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}$$

where A_x , A_y , and A_z are the scalar components, and **i**, **j**, **k** are the standard unit vectors along the x, y, and z axes respectively.

Key Properties

Direction

Indicated by the arrowhead pointing from tail to head.

Magnitude

Proportional to the length of the arrow, drawn to scale.

Equality

$\overrightarrow{AB} = \overrightarrow{CD}$ means $\vec{AB}$ same magnitude and same direction.

Vector addition, subtraction and scalar multiplication of vectors:

Question : Suppose, $\underline{A} = 4\hat{i} - 3\hat{j} + 2\hat{k}$, $\underline{B} = 5\hat{i} + 4\hat{j} - 5\hat{k}$ and $\underline{C} = 2\hat{i} - 4\hat{j} - 2\hat{k}$ are three vectors. Find of $\underline{A} + \underline{B} + \underline{C}$, $3\underline{B}$, $5\underline{C}$, $3\vec{A} - 2\vec{B} + 4\vec{C}$

Solution: $\underline{A} + \underline{B} + \underline{C} = 4\hat{i} - 3\hat{j} + 2\hat{k} + 5\hat{i} + 4\hat{j} - 5\hat{k} + 2\hat{i} - 4\hat{j} - 2\hat{k} = 11\hat{i} - 3\hat{j} - 5\hat{k}$

And, $3\underline{B} = 3.(5\hat{i} + 4\hat{j} - 5\hat{k})$

$5\underline{C} = 5.(2\hat{i} - 4\hat{j} - 2\hat{k})$

And, $3\vec{A} - 2\vec{B} + 4\vec{C} = 3.(4\hat{i} - 3\hat{j} + 2\hat{k}) - 2.(5\hat{i} + 4\hat{j} - 5\hat{k}) + 4.(2\hat{i} - 4\hat{j} - 2\hat{k})$
 $= 12\hat{i} - 9\hat{j} + 6\hat{k} - 10\hat{i} + 8\hat{j} + 10\hat{k} + 8\hat{i} - 16\hat{j} - 8\hat{k}$
 $= 10\hat{i} - 17\hat{j} + 8\hat{k}$

Magnitude of a vector:

If $\vec{A} = a\hat{i} + b\hat{j} + c\hat{k}$ be a vector then the magnitude of $\vec{A}$ is denoted by A or $|\vec{A}|$ and is defined as follows:

$$A = |\vec{A}| = \sqrt{a^2 + b^2 + c^2}$$

Example 1: If $\vec{B} = 2\hat{i} + 4\hat{j} - 5\hat{k}$, then the magnitude of $\vec{B}$ is,

$$B = |\vec{B}| = \sqrt{2^2 + 4^2 + (-5)^2} = \sqrt{4 + 16 + 25} = \sqrt{45}$$

Example 2: If $\vec{A} = 4\hat{i} - 3\hat{j} + 2\hat{k}$, $\vec{B} = 5\hat{i} - 4\hat{j} + 3\hat{k}$, $\vec{C} = 2\hat{i} - 4\hat{j} + 2\hat{k}$

Then $\vec{A} - 2\vec{B} + \vec{C} = 4\hat{i} - 3\hat{j} + 2\hat{k} - 2(5\hat{i} + 4\hat{j} - 5\hat{k}) + 2\hat{i} - 4\hat{j} - 2\hat{k}$.

$$= 4\hat{i} - 3\hat{j} + 2\hat{k} - 10\hat{i} - 8\hat{j} + 10\hat{k} + 2\hat{i} - 4\hat{j} - 2\hat{k}.$$

$$= -4\hat{i} - 15\hat{j} + 10\hat{k}$$

And $|\vec{A} - 2\vec{B} + \vec{C}| = \sqrt{(-4)^2 + (-15)^2 + 10^2} = \sqrt{341}$.

Question 3: Suppose, $\underline{A} = 4\hat{i} - 3\hat{j} + 2\hat{k}$, $\underline{B} = 5\hat{i} + 4\hat{j} - 5\hat{k}$ and $\underline{C} = 2\hat{i} - 4\hat{j} - 2\hat{k}$ are three vectors. Find the magnitude of $\underline{A} + \underline{B} + \underline{C}$ and $(3\underline{B} - 5\underline{C})$.

Solution: $\underline{A} + \underline{B} + \underline{C} = 4\hat{i} - 3\hat{j} + 2\hat{k} + 5\hat{i} + 4\hat{j} - 5\hat{k} + 2\hat{i} - 4\hat{j} - 2\hat{k} = 11\hat{i} - 3\hat{j} - 5\hat{k}$

Now, $|\underline{A} + \underline{B} + \underline{C}| = \sqrt{(11)^2 + (-3)^2 + (-5)^2} = \sqrt{155}$

And, $3\underline{B} - 5\underline{C} = 3.(5\hat{i} + 4\hat{j} - 5\hat{k}) - 5.(2\hat{i} - 4\hat{j} - 2\hat{k})$
 $= 15\hat{i} + 12\hat{j} - 15\hat{k} - 10\hat{i} + 20\hat{j} + 10\hat{k}$
 $= 5\hat{i} + 32\hat{j} - 5\hat{k}$

The magnitude of $(3\underline{B} - 5\underline{C})$ is,

$$|3\underline{B} - 5\underline{C}| = \sqrt{5^2 + 32^2 + (-5)^2} = \sqrt{25 + 1024 + 25} = \sqrt{1074}$$

Vector product

There are **two types** of vector product. (i) **Dot product**, (ii) **Cross product**

Dot product: If $\underline{A} = a\hat{i} + b\hat{j} + c\hat{k}$ and $\underline{B} = h\hat{i} + l\hat{j} + m\hat{k}$ are two vectors then the dot product of $\underline{A}$ and $\underline{B}$ is denoted by $\underline{A} \cdot \underline{B}$ and is defined as follows:

$$\underline{A} \cdot \underline{B} = (a\hat{i} + b\hat{j} + c\hat{k}) \cdot (h\hat{i} + l\hat{j} + m\hat{k}) = ah + bl + cm$$

Since, $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$ and $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$

Finding the Angle

If $\underline{A}$ and $\underline{B}$ are two vectors and θ be the angle between them then

$$\underline{A} \cdot \underline{B} = |\underline{A}| |\underline{B}| \cos \theta$$

$$\text{or, } |\underline{A}| |\underline{B}| \cos \theta = \underline{A} \cdot \underline{B}$$

$$\text{or, } \cos \theta = \frac{\underline{A} \cdot \underline{B}}{|\underline{A}| |\underline{B}|}$$

$$\text{or, } \theta = \cos^{-1} \left(\frac{\underline{A} \cdot \underline{B}}{|\underline{A}| |\underline{B}|} \right)$$

Question: Suppose, $\underline{A} = 4\hat{i} - 3\hat{j} + 2\hat{k}$, $\underline{B} = 5\hat{i} + 4\hat{j} - 5\hat{k}$ and $\underline{C} = 2\hat{i} - 4\hat{j} - 2\hat{k}$, then show that $(\underline{A} + \underline{B}).(\underline{A} - \underline{B}) = |\underline{A}|^2 - |\underline{B}|^2$

Solution: $\underline{A} + \underline{B} = 9\hat{i} + \hat{j} - 3\hat{k}$, $(\underline{A} - \underline{B}) = -\hat{i} - 7\hat{j} + 7\hat{k}$

$$\text{L.H.S} = (\underline{A} + \underline{B}).(\underline{A} - \underline{B}) = (9\hat{i} + \hat{j} - 3\hat{k})(-\hat{i} - 7\hat{j} + 7\hat{k}) = -9 - 7 - 21 = -37$$

$$\text{Again } |\underline{A}| = \sqrt{29} \text{ and } |\underline{B}| = \sqrt{5^2 + 4^2 + (-5)^2} = \sqrt{66}$$

$$\text{R.H.S} = |\underline{A}|^2 - |\underline{B}|^2 = 29 - 66 = -37$$

Question: Suppose, $\underline{A} = 4\hat{i} - 3\hat{j} + 2\hat{k}$, $\underline{B} = 5\hat{i} + 4\hat{j} - 5\hat{k}$ and $\underline{C} = 2\hat{i} - 4\hat{j} - 2\hat{k}$, then find the value of $(\underline{A} + \underline{B}).(\underline{B} - \underline{C})$

$$\textbf{Solution: } \underline{A} + \underline{B} = 4\hat{i} - 3\hat{j} + 2\hat{k} + 5\hat{i} + 4\hat{j} - 5\hat{k} = 9\hat{i} + \hat{j} - 3\hat{k}$$

$$(\underline{B} - \underline{C}) = 5\hat{i} + 4\hat{j} - 5\hat{k} - 2\hat{i} + 4\hat{j} + 2\hat{k} = 3\hat{i} + 8\hat{j} - 3\hat{k}$$

$$\text{So, } (\underline{A} + \underline{B}).(\underline{B} - \underline{C}) = (9\hat{i} + \hat{j} - 3\hat{k}).(3\hat{i} + 8\hat{j} - 3\hat{k})$$

$$= 27 + 8 + 9 = 44$$

Question: Suppose, $\underline{A} = 4\hat{i} - 3\hat{j} + 2\hat{k}$, $\underline{B} = 5\hat{i} + 4\hat{j} - 5\hat{k}$ and $\underline{C} = 2\hat{i} - 4\hat{j} - 2\hat{k}$, then Find the angle between $\underline{A}$ and $\underline{B}$

Solution: Let θ be the angle between $\underline{A}$ and $\underline{B}$.

We know that, $\underline{A} \cdot \underline{B} = |\underline{A}| |\underline{B}| \cos \theta$

$$\therefore, \theta = \cos^{-1} \left(\frac{\underline{A} \cdot \underline{B}}{|\underline{A}| |\underline{B}|} \right) \dots \dots \dots (i)$$

Now, $\underline{A} \cdot \underline{B} = (4\hat{i} - 3\hat{j} + 2\hat{k}) \cdot (5\hat{i} + 4\hat{j} - 5\hat{k}) = 20 - 12 - 10 = 20 - 22 = -2$

$|\underline{A}| = \sqrt{29}$ & $|\underline{B}| = \sqrt{66}$

From (i) we get, $\theta = \cos^{-1} \left(\frac{-2}{\sqrt{29} \cdot \sqrt{66}} \right)$

Exercise: Suppose, $\underline{A} = 4\hat{i} - 3\hat{j} + 2\hat{k}$, $\underline{B} = 5\hat{i} + 4\hat{j} - 5\hat{k}$ and $\underline{C} = 2\hat{i} - 4\hat{j} - 2\hat{k}$

(i) Show that $(\underline{A} - \underline{B}) \cdot \underline{C} = \underline{A} \cdot \underline{C} - \underline{B} \cdot \underline{C}$

(ii) Show that $(\underline{A} + \underline{B}) \cdot \underline{C} = \underline{A} \cdot \underline{C} + \underline{B} \cdot \underline{C}$

(iii) Find the angle between $\underline{B}$ and $\underline{C}$

Cross Product (Vector Product)

Definition and Computation

If $\underline{A} = a\hat{i} + b\hat{j} + c\hat{k}$ and $\underline{B} = h\hat{i} + l\hat{j} + m\hat{k}$ are two vectors then the cross product of $\underline{A}$ and $\underline{B}$ is denoted by $\underline{A} \times \underline{B}$ and is defined as follows:

$$\underline{A} \times \underline{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a & b & c \\ h & l & m \end{vmatrix} = (bm - cl)\hat{i} - (am - ch)\hat{j} + (al - bh)\hat{k}$$

Angle: Again, If $\underline{A}$ and $\underline{B}$ are two vectors and θ be the angle between them then

$$\underline{A} \times \underline{B} = |\underline{A}||\underline{B}|\sin \theta$$

Example: If $\underline{A} = 4\hat{i} - 3\hat{j} + 2\hat{k}$ and $\underline{B} = 5\hat{i} + 4\hat{j} - 5\hat{k}$ are two vectors then find $(\underline{A} \times \underline{B})$.

$$\textbf{Solution: } (\underline{A} \times \underline{B}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -3 & 2 \\ 5 & 4 & -5 \end{vmatrix} = 7\hat{i} + 30\hat{j} + 31\hat{k}$$

$$(\underline{B} \times \underline{A}) = -7\hat{i} - 30\hat{j} - 31\hat{k} = -(7\hat{i} + 30\hat{j} + 31\hat{k}) = -(\underline{A} \times \underline{B})$$

Unit Vector

If $\underline{A}$ be any vector and $|\underline{A}|$ be the magnitude of $\underline{A}$ then the unit vector along $\underline{A}$ is denoted by $\hat{a}$ and is defined as follows $\hat{a} = \frac{\underline{A}}{|\underline{A}|}$; $|\hat{a}| = 1$.

The unit vector along $(\underline{A} \times \underline{B})$ is $= \frac{(\underline{A} \times \underline{B})}{|\underline{A} \times \underline{B}|}$

The unit vector has magnitude exactly 1 and points in the same direction as $\underline{A}$.

Worked Example

Question 1: If $\underline{A} = 4\hat{i} - 3\hat{j} + 2\hat{k}$ and $\underline{B} = 5\hat{i} - 4\hat{j} - 5\hat{k}$ then find the unit vector along $(\underline{A} + \underline{B})$.

Solution: Given, $\underline{A} = 4\hat{i} - 3\hat{j} + 2\hat{k}$ and $\underline{B} = 5\hat{i} - 4\hat{j} - 5\hat{k}$

$$\text{Now } \underline{A} + \underline{B} = 4\hat{i} - 3\hat{j} + 2\hat{k} + 5\hat{i} - 4\hat{j} - 5\hat{k} = 9\hat{i} - \hat{j} - 3\hat{k}$$

$$\text{And } |\underline{A} + \underline{B}| = \sqrt{9^2 + (-1)^2 + (-3)^2} = \sqrt{81 + 1 + 9} = \sqrt{91}$$

$$\text{So, the unit vector along } (\underline{A} + \underline{B}) \text{ is } = \frac{(\underline{A} + \underline{B})}{|\underline{A} + \underline{B}|} = \frac{9\hat{i} - \hat{j} - 3\hat{k}}{\sqrt{91}} = \frac{9}{\sqrt{91}}\hat{i} - \frac{1}{\sqrt{91}}\hat{j} - \frac{3}{\sqrt{91}}\hat{k}$$

Question 2: If $\underline{P} = 4\hat{i} - 2\hat{j} + 3\hat{k}$, $\underline{Q} = 7\hat{i} + 2\hat{j} - 5\hat{k}$ and $\underline{R} = 3\hat{i} - 4\hat{j} - \hat{k}$.

Then, find the unit vector along $(\underline{P} \times \underline{Q})$, $(\underline{P} + \underline{Q} - \underline{R})$, $(\underline{P} - 2\underline{Q} - 3\underline{R})$ and $(\underline{P} + \underline{Q} + \underline{R})$

Solution: Given, $\underline{P} = 4\hat{i} - 2\hat{j} + 3\hat{k}$, $\underline{Q} = 7\hat{i} + 2\hat{j} - 5\hat{k}$ and $\underline{R} = 3\hat{i} - 4\hat{j} - \hat{k}$.

$$\begin{aligned}\text{Now } \underline{P} + \underline{Q} - \underline{R} &= 4\hat{i} - 2\hat{j} + 3\hat{k} + 7\hat{i} + 2\hat{j} - 5\hat{k} - 3\hat{i} - 4\hat{j} - \hat{k} \\ &= 8\hat{i} - 4\hat{j} - 3\hat{k}\end{aligned}$$

$$\text{And } |\underline{P} + \underline{Q} - \underline{R}| = \sqrt{8^2 + (-4)^2 + (-3)^2} = \sqrt{64 + 16 + 9} = \sqrt{89}$$

$$\text{So, the unit vector along } (\underline{P} + \underline{Q} - \underline{R}) \text{ is } = \frac{(\underline{P} + \underline{Q} - \underline{R})}{|\underline{P} + \underline{Q} - \underline{R}|} = \frac{8\hat{i} - 4\hat{j} - 3\hat{k}}{\sqrt{89}} = \frac{8}{\sqrt{89}}\hat{i} - \frac{4}{\sqrt{89}}\hat{j} - \frac{3}{\sqrt{89}}\hat{k}$$

Exercise: If $\underline{P} = 4\hat{i} - 2\hat{j} + 3\hat{k}$, $\underline{Q} = 7\hat{i} + 2\hat{j} - 5\hat{k}$ and $\underline{R} = 3\hat{i} - 4\hat{j} - \hat{k}$.

Then, find the unit vector along $(\underline{P} \times \underline{Q})$, $(\underline{P} - 2\underline{Q} - 3\underline{R})$ and $(\underline{P} + \underline{Q} + \underline{R})$